

# CDK12 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes CDK12 gene fusions from the msk\_impact\_50k\_2026 study, covering 43 fusion events. 7/43 fusions (16.3%) were in-frame. 16/43 fusions (37.2%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ( $p=0.0677006$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for CDK12.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 43 analyzed fusion events, identifying 29 distinct fusion partner genes. The most frequent partner was FBXL20, observed in 6 events.

### Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain,  $p=0.0677006$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.316832.

### Domain disruption

Domain-disruption analysis was skipped: CDK12 has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 41 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 644 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is -84 aa from the nearest configured domain boundary at 728 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=16$ ) and "not\_retained" ( $n=25$ ). For Frame\_status, retained: 5/7 (71.4%, 95% CI 35.9%-91.8%); not\_retained: 2/7 (28.6%, 95% CI 8.2%-64.1%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 20.8125 vs 43.4,  $p=0.0502833$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: CDK12 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| ADCY9        | 2           | 0.046511627906976744 | 0.04991713954627366    | None                    | 1.0                      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2756851385529113  | 1    |
| AKAP8L       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.9906291834002677       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2652071802079033  | 2    |
| HTR1E        | 2           | 0.046511627906976744 | 0.04991713954627366    | None                    | 0.9250334672021419       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2616289136533129  | 3    |
| MED13        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.9451137884872824       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.25667304366171856 | 4    |
| ODAD4        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.9210174029451138       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.252154971372562   | 5    |
| TBKBP1       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.9210174029451138       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.252154971372562   | 6    |
| GRB7         | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.85809906291834         | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.24035778261754187 | 7    |
| ERBB2        | 3           | 0.06976744186046512  | 0.04991713954627366    | None                    | 0.7367246764837126       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.23504194562616557 | 8    |
| RECQL4       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.7817938420348058       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.22605055370187926 | 9    |
| STAT5B       | 4           | 0.09302325581395349  | 0.04991713954627366    | None                    | 0.5917001338688086       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21657077411842918 | 10   |
| COLEC12      | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.6974564926372155       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21023730068983107 | 11   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| CACNA1G      | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.6613119143239625       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20346019225609613 | 12   |
| FBXL20       | 6           | 0.13953488372093023  | 0.04991713954627366    | None                    | 0.4152164212405176       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20092193846574088 | 13   |
| SYNRG        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.6265060240963856       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.19693408783842545 | 14   |
| ANKFN1       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.6050870147255689       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1929180235813973  | 15   |
| SKAP1        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.5917001338688086       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.19040798342075477 | 16   |
| USP6         | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.5917001338688086       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.19040798342075477 | 17   |
| CSF3         | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.5394912985274432       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.18061882679424873 | 18   |
| FAM222B      | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.4029451137884873       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.15501641715569453 | 19   |
| IKZF3        | 2           | 0.046511627906976744 | 0.04991713954627366    | None                    | 0.321285140562249        | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.14842610240833298 | 20   |
| SPON1        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.32797858099062915      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1409601922560961  | 21   |
| SEC31A       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.2449799196787149       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1253979432601122  | 22   |
| VMP1         | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.2449799196787149       | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1253979432601122  | 23   |
| SNF8         | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.18340026773761708      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.11385175852115635 | 24   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| MAP3K3       | 2           | 0.046511627906976744 | 0.04991713954627366    | None                    | None                     | 0.441153375836676          | ['confidence_certainty', 'domain_retention', 'recurrence']                       | 0.10853555514204467 | 25   |
| MTRES1       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.048192771084337394     | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08850035289866641 | 26   |
| IGFBP4       | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.01740294511378848      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08272726052918851 | 27   |
| FGFR4        | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.01338688085676043      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08197424848099574 | 28   |
| RPS6KB1      | 1           | 0.023255813953488372 | 0.04991713954627366    | None                    | 0.0                      | 0.441153375836676          | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.07946420832035316 | 29   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio         | p_value               | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|--------------------|-----------------------|---------------------|
| 80       | 1                      | 15               | 0                      | 25               | None               | 0.39024390243902435   | 0.4086638740638108  |
| 240      | 2                      | 14               | 0                      | 25               | None               | 0.14634146341463414   | 0.8346326063360919  |
| 246      | 3                      | 13               | 0                      | 25               | None               | 0.052532833020637895  | 1.279569177684353   |
| 349      | 4                      | 12               | 0                      | 25               | None               | 0.01797175866495507   | 1.7454094219943264  |
| 603      | 5                      | 11               | 0                      | 25               | None               | 0.0058286784859313745 | 2.2344299000136965  |
| 637      | 6                      | 10               | 0                      | 25               | None               | 0.0017809850929234755 | 2.749339715622759   |
| 644      | 9                      | 7                | 0                      | 25               | None               | 3.265366098560993e-05 | 4.486068120461908   |
| 700      | 9                      | 7                | 2                      | 23               | 14.785714285714286 | 0.0011510086327457892 | 2.938921419081068   |
| 703      | 11                     | 5                | 2                      | 23               | 25.3               | 7.698377534159563e-05 | 4.113600794594795   |
| 750      | 11                     | 5                | 3                      | 22               | 16.133333333333333 | 0.0004274646628148892 | 3.369099781237133   |
| 807      | 11                     | 5                | 4                      | 21               | 11.55              | 0.0009913088581971262 | 3.003791013009668   |
| 870      | 11                     | 5                | 5                      | 20               | 8.8                | 0.003014843556286022  | 2.520735218981372   |
| 897      | 11                     | 5                | 6                      | 19               | 6.966666666666667  | 0.008600185164931898  | 2.0654921981468433  |
| 923      | 11                     | 5                | 9                      | 16               | 3.911111111111111  | 0.05784571299099636   | 1.2377288215582707  |
| 949      | 11                     | 5                | 15                     | 10               | 1.4666666666666666 | 0.7420848641039097    | 0.12954642638599925 |
| 955      | 11                     | 5                | 16                     | 9                | 1.2375             | 1.0                   | -0.0                |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio         | p_value             | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|--------------------|---------------------|---------------------|
| 988      | 11                     | 5                | 17                     | 8                | 1.035294117647059  | 1.0                 | -0.0                |
| 1032     | 12                     | 4                | 17                     | 8                | 1.411764705882353  | 0.7341702326198091  | 0.13420322820990868 |
| 1090     | 12                     | 4                | 18                     | 7                | 1.166666666666667  | 1.0                 | -0.0                |
| 1121     | 12                     | 4                | 19                     | 6                | 0.9473684210526315 | 1.0                 | -0.0                |
| 1146     | 13                     | 3                | 19                     | 6                | 1.368421052631579  | 1.0                 | -0.0                |
| 1159     | 13                     | 3                | 20                     | 5                | 1.083333333333333  | 1.0                 | -0.0                |
| 1208     | 13                     | 3                | 22                     | 3                | 0.5909090909090909 | 0.6623956829348356  | 0.17888250666352729 |
| 1254     | 15                     | 1                | 22                     | 3                | 2.0454545454545454 | 1.0                 | -0.0                |
| 1355     | 16                     | 0                | 22                     | 3                | None               | 0.26829268292682923 | 0.5713911715615105  |
| 1378     | 16                     | 0                | 23                     | 2                | None               | 0.5121951219512195  | 0.2905645619858162  |
| 1381     | 16                     | 0                | 24                     | 1                | None               | 1.0                 | -0.0                |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|   |    |
|---|----|
| 5 | 11 |
| 2 | 23 |

*domain\_retention\_descriptives*

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 41                      | 16                   | 15              | 10               | 0.22484641638225256                             |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 4    | 19                   | 43                | 0.4418604651162791 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| ADCY9        | 2           |
| AKAP8L       | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| ANKFN1       | 1           |
| CACNA1G      | 1           |
| COLEC12      | 1           |
| CSF3         | 1           |
| ERBB2        | 3           |
| FAM222B      | 1           |
| FBXL20       | 6           |
| FGFR4        | 1           |
| GRB7         | 1           |
| HTR1E        | 2           |
| IGFBP4       | 1           |
| IKZF3        | 2           |
| MAP3K3       | 2           |
| MED13        | 1           |
| MTRES1       | 1           |
| ODAD4        | 1           |
| RECQL4       | 1           |
| RPS6KB1      | 1           |
| SEC31A       | 1           |
| SKAP1        | 1           |
| SNF8         | 1           |
| SPON1        | 1           |
| STAT5B       | 4           |
| SYNRG        | 1           |
| TBKBP1       | 1           |
| USP6         | 1           |
| VMP1         | 1           |

**Per-event results (results.tsv)**

| event_id                         | sample_id                 | patient_id | fusion_name           | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                | disrupted_domains           | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>9125-T01-I<br>M6-1  | P-0039<br>125-T0<br>1-IM6 |            | AKAP<br>8L-CD<br>K12  | AKA<br>P8L   | unknown      | three_prime | 376279<br>94       | 2               | 637                         | False                | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |                             |                             | AKAP8L (NM_014371) -<br>CDK12 (NM_016507)<br>rearrangement: t(17;19)(q12;p13.12)(chr17:g.37627994::chr19:g.15529663)  <br>Protein Fusion: mid-exon<br>{AKAP8L:CDK12} | NA                           |
| EVT-P-005<br>6622-T01-I<br>M6-2  | P-0056<br>622-T0<br>1-IM6 |            | ANKF<br>N1-CD<br>K12  | ANK<br>FN1   | in-frame     | three_prime | 376193<br>99       | 1               | 349                         | True                 | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |                             |                             | ANKFN1 (NM_153228) -<br>CDK12 (NM_016507)<br>rearrangement: c.62+8631:<br>ANKFN1_c.1046+29:CDK12dup  <br>Protein Fusion: in frame<br>{ANKFN1:CDK12}                  | NA                           |
| EVT-P-000<br>3762-T02-I<br>M5-4  | P-0003<br>762-T0<br>2-IM5 |            | CDK1<br>2-ADC<br>Y9   | ADC<br>Y9    | unknown      | three_prime | 376466<br>65       | 3               | 644                         | True                 | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |                             |                             | ADCY9 (NM_001116) -<br>CDK12 (NM_016507)<br>rearrangement: t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.37646665)  <br>Protein fusion: mid-exon<br>(ADCY9-CDK12)     | NA                           |
| EVT-P-000<br>3762-T01-I<br>M5-5  | P-0003<br>762-T0<br>1-IM5 |            | CDK1<br>2-ADC<br>Y9   | ADC<br>Y9    | unknown      | three_prime | 376466<br>65       | 3               | 644                         | True                 | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |                             |                             | t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.37646665)  <br>Protein fusion: mid-exon<br>(ADCY9-CDK12)                                                                | NA                           |
| EVT-P-002<br>4871-T01-I<br>M6-7  | P-0024<br>871-T0<br>1-IM6 |            | CDK1<br>2-CAC<br>NA1G | CAC<br>NA1G  | unknown      | five_prime  | 376678<br>06       | 8               | 897                         | False                | disrupted     | 0.5802047<br>781569966   | True                |                             |                             | Protein<br>kinase<br>domain | CDK12 (NM_016507) -<br>CACNA1G (NM_018896)<br>rearrangement: c.2691:CDK12_c.6399+226:CACNA1Gdel  <br>Protein Fusion: mid-exon<br>{CDK12:CACNA1G}                     | NA                           |
| EVT-P-003<br>8990-T01-I<br>M6-53 | P-0038<br>990-T0<br>1-IM6 |            | CDK1<br>2-FBX<br>L20  | FBX<br>L20   | in-frame     | five_prime  | 376763<br>67       | 11              | 1032                        | True                 | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |                             |                             | CDK12 (NM_016507) -<br>FBXL20 (NM_032875)<br>rearrangement: c.3095+27:<br>CDK12_c.104+18457:FBXL20inv  <br>Protein Fusion: in frame<br>{CDK12:FBXL20}                | NA                           |
| EVT-P-000<br>2174-T01-I<br>M3-55 | P-0002<br>174-T0<br>1-IM3 |            | CDK1<br>2-GRB<br>7    | GR<br>B7     | unknown      | three_prime | 376507<br>68       | 5               | 750                         | True                 | disrupted     | 0.9249146<br>757679181   | True                |                             |                             | Protein<br>kinase<br>domain | NA   Transcript fusion<br>(GRB7-CDK12)                                                                                                                               | NA                           |
| EVT-P-000<br>8960-T02-I<br>M6-56 | P-0008<br>960-T0<br>2-IM6 |            | CDK1<br>2-HTR<br>1E   | HTR<br>1E    | unknown      | five_prime  | 376469<br>77       | 3               | 700                         | False                | lost          | 0.0                      | False               |                             | Protein<br>kinase<br>domain |                             | CDK12 (NM_016507) -<br>HTR1E (NM_000865)<br>rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697113::chr17:g.37646977)  <br>Protein Fusion: mid-exon<br>{CDK12:HTR1E}      | NA                           |



| event_id                  | sample_id         | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains     | source_annotation_text                                                                                                                                     | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008960-T01-I M5-57 | P-0008960-T01-IM5 |            | CDK12-HTR1E   | HTR1E        | unknown      | five_prime  | 37646977           | 3               | 700                         | False                | lost          | 0.0                      | False               |                       | Protein kinase domain |                       | CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697112::chr17:g.37646977)   Protein fusion: mid-exon {CDK12-HTR1E}        | NA                           |
| EVT-P-0060111-T01-I M7-58 | P-0060111-T01-IM7 |            | CDK12-IKZF3   | IKZF3        | in-frame     | five_prime  | 37682742           | 13              | 1254                        | True                 | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | CDK12 (NM_016507) - IKZF3 (NM_012481) rearrangement: c.3760+173:CDK12_c.8-12056:IKZF3inv   Protein Fusion: in frame {CDK12:IKZF3}                          | NA                           |
| EVT-P-0040752-T01-I M6-62 | P-0040752-T01-IM6 |            | CDK12-MTR ES1 | MTR ES1      | unknown      | five_prime  | 37687160           | 14              | 1355                        | False                | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | CDK12 (NM_016507) - C6orf203 (NM_016487) rearrangement: t(6;17)(q21;q12)(chr6:g.107368450::chr17:g.37687160)   Protein Fusion: mid-exon {CDK12:C6orf203}   | NA                           |
| EVT-P-0009835-T01-I M5-67 | P-0009835-T01-IM5 |            | CDK12-RECQL4  | RECQL4       | out-of-frame | five_prime  | 37656296           | 6               | 807                         | True                 | disrupted     | 0.27303754266211605      | True                |                       |                       | Protein kinase domain | CDK12 (NM_016507) - RECQL4 (NM_004260) rearrangement: t(8;17)(q24.3;q12)(chr8:g.145739493::chr17:g.37656296)   Protein fusion: out of frame {CDK12-RECQL4} | NA                           |
| EVT-P-0023550-T01-I M6-68 | P-0023550-T01-IM6 |            | CDK12-RPS6KB1 | RPS6KB1      | unknown      | three_prime | 37687269           | 14              | 1391                        | False                | lost          | 0.0                      | False               |                       | Protein kinase domain |                       | CDK12 (NM_016507) rearrangement: c.4173_chr17:g.58028299dup   Protein Fusion: mid-exon {RPS6KB1:CDK12}                                                     | NA                           |
| EVT-P-0041371-T01-I M6-69 | P-0041371-T01-IM6 |            | CDK12-SKAP1   | SKAP1        | out-of-frame | five_prime  | 37673680           | 10              | 949                         | True                 | disrupted     | 0.757679180887372        | True                |                       |                       | Protein kinase domain | CDK12 (NM_016507) - SKAP1 (NM_003726) rearrangement: c.2847-13:CDK12_c.280+66555:SKAP1inv   Protein Fusion: out of frame {CDK12:SKAP1}                     | NA                           |
| EVT-P-0043326-T01-I M6-71 | P-0043326-T01-IM6 |            | CDK12-SNF8    | SNF8         | out-of-frame | five_prime  | 37683411           | 13              | 1254                        | True                 | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | CDK12 (NM_016507) - SNF8 (NM_007241) rearrangement: c.3760+842:CDK12_c.422+1405:SNF8inv   Protein Fusion: out of frame {CDK12:SNF8}                        | NA                           |
| EVT-P-0057341-T01-I M6-73 | P-0057341-T01-IM6 |            | CDK12-SPON1   | SPON1        | unknown      | five_prime  | 37682247           | 13              | 1146                        | False                | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | CDK12 (NM_016507) - SPON1 (NM_006108) rearrangement: t(11;17)(p15.2;q12)(chr11:g.14149386::chr17:g.37682247)   Protein Fusion: mid-exon {CDK12:SPON1}      | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name           | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction    | domain_is_truncated | retained_domains            | lost_domains | disrupted_domains           | source_annotation_text                                                                                                                                                          | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------------|---------------------|-----------------------------|--------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6507-T04-I<br>M6-74 | P-0016<br>507-T0<br>4-IM6 |            | CDK1<br>2-STA<br>T5B  | STAT5B       | unknown      | three_prime | 37672113           | 9               | 949                         | True                   | disrupted     | 0.2457337<br>883959044<br>3 | True                |                             |              | Protein<br>kinase<br>domain | CDK12 (NM_016507) -<br>STAT5B (NM_012448)<br>rearrangement: c.2846+52<br>:CDK12_c.2238-225:STAT<br>5Bdel   Antisense Fusion                                                     | NA                           |
| EVT-P-001<br>6507-T03-I<br>M6-75 | P-0016<br>507-T0<br>3-IM6 |            | CDK1<br>2-STA<br>T5B  | STAT5B       | unknown      | three_prime | 37672113           | 9               | 949                         | True                   | disrupted     | 0.2457337<br>883959044<br>3 | True                |                             |              | Protein<br>kinase<br>domain | CDK12 (NM_016507) -<br>STAT5B (NM_012448)<br>rearrangement: c.2846+52<br>:CDK12_c.2238-225:STAT<br>5Bdel.   Antisense Fusion                                                    | NA                           |
| EVT-P-001<br>6507-T02-I<br>M6-76 | P-0016<br>507-T0<br>2-IM6 |            | CDK1<br>2-STA<br>T5B  | STAT5B       | unknown      | three_prime | 37672113           | 9               | 949                         | True                   | disrupted     | 0.2457337<br>883959044<br>3 | True                |                             |              | Protein<br>kinase<br>domain | CDK12 (NM_016507) -<br>STAT5B (NM_012448)<br>rearrangement: c.2846+52<br>:CDK12_c.2238-225:STAT<br>5Bdel   Antisense Fusion                                                     | NA                           |
| EVT-P-001<br>6507-T01-I<br>M6-77 | P-0016<br>507-T0<br>1-IM6 |            | CDK1<br>2-STA<br>T5B  | STAT5B       | unknown      | three_prime | 37672113           | 9               | 949                         | True                   | disrupted     | 0.2457337<br>883959044<br>3 | True                |                             |              | Protein<br>kinase<br>domain | CDK12 (NM_016507) -<br>STAT5B (NM_012448)<br>rearrangement: c.2846+52<br>:CDK12_c.2238-225:STAT<br>5Bdel   Antisense Fusion                                                     | NA                           |
| EVT-P-002<br>3516-T01-I<br>M6-78 | P-0023<br>516-T0<br>1-IM6 |            | COLE<br>C12-C<br>DK12 | COLEC12      | out-of-frame | three_prime | 37665943           | 7               | 870                         | True                   | disrupted     | 0.5153583<br>61774744       | True                |                             |              | Protein<br>kinase<br>domain | COLEC12 (NM_130386) -<br>CDK12(NM_016507)<br>rearrangement: t(17;18)(q1<br>2;p11.32)(chr17:g.376659<br>43::chr18:g.386902)  <br>Protein Fusion: out of<br>frame {COLEC12:CDK12} | NA                           |
| EVT-P-003<br>8406-T01-I<br>M6-80 | P-0038<br>406-T0<br>1-IM6 |            | CSF3-<br>CDK1<br>2    | CSF3         | out-of-frame | three_prime | 37676083           | 11              | 988                         | True                   | disrupted     | 0.1126279<br>863481228<br>6 | True                |                             |              | Protein<br>kinase<br>domain | CSF3 (NM_000759) -<br>CDK12 (NM_016507)<br>Rearrangement : c.313-58:<br>CSF3_c.2964-126:CDK12<br>dup   Protein Fusion: out of<br>frame {CSF3:CDK12}                             | NA                           |
| EVT-P-004<br>6790-T01-I<br>M6-81 | P-0046<br>790-T0<br>1-IM6 |            | ERBB<br>2-CDK<br>12   | ERBB2        | out-of-frame | three_prime | 37640037           | 3               | 644                         | True                   | retained      | 1.0                         | False               | Protein<br>kinase<br>domain |              |                             | ERBB2 (NM_004448) -<br>CDK12 (NM_016507)<br>rearrangement: c.225+207<br>:ERBB2_c.1932-6773:CD<br>K12dup   Protein Fusion:<br>out of frame<br>{ERBB2:CDK12}                      | NA                           |
| EVT-P-004<br>3706-T01-I<br>M6-82 | P-0043<br>706-T0<br>1-IM6 |            | ERBB<br>2-CDK<br>12   | ERBB2        | unknown      | three_prime | 37668120           | 8               | 923                         | True                   | disrupted     | 0.3344709<br>897610921<br>3 | True                |                             |              | Protein<br>kinase<br>domain | ERBB2 (NM_004448) -<br>CDK12 (NM_016507)<br>rearrangement: c.2293:ER<br>BB2_c.2768+237:CDK12d<br>up   Protein Fusion:<br>mid-exon {ERBB2:CDK12}                                 | NA                           |
| EVT-P-003<br>7167-T01-I<br>M6-83 | P-0037<br>167-T0<br>1-IM6 |            | ERBB<br>2-CDK<br>12   | ERBB2        | unknown      | three_prime | 37673710           | 10              | 955                         | False                  | disrupted     | 0.2252559<br>726962457<br>3 | True                |                             |              | Protein<br>kinase<br>domain | ERBB2 (NM_004448) -<br>CDK12 (NM_016507)<br>rearrangement: c.2553:ER<br>BB2_c.2864:CDK12dup  <br>Protein Fusion: mid-exon<br>{ERBB2:CDK12}                                      | NA                           |

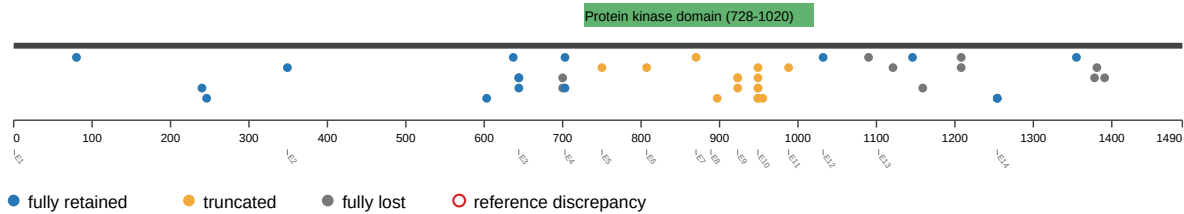
| event_id                         | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role             | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction    | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains     | source_annotation_text                                                                                                                              | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------------|---------------------|-----------------------|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>6357-T01-I<br>M6-84 | P-0036<br>357-T0<br>1-IM6 |            | FAM2<br>22B-CDK12 | FAM<br>22B   | unknown      | thr<br>ee<br>_pr<br>ime | 376811<br>00       | 12              | 1090                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | CDK12 (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv   Protein Fusion: mid-exon {FAM22B:CDK12}                                         | NA                           |
| EVT-P-001<br>4403-T01-I<br>M6-86 | P-0014<br>403-T0<br>1-IM6 |            | FBXL2<br>0-CDK12  | FBXL20       | unknown      | thr<br>ee<br>_pr<br>ime | 376821<br>71       | 13              | 1121                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv   Protein fusion: mid-exon (FBXL20-CDK12)                    | NA                           |
| EVT-P-004<br>6675-T01-I<br>M6-87 | P-0046<br>675-T0<br>1-IM6 |            | FBXL2<br>0-CDK12  | FBXL20       | in-frame     | thr<br>ee<br>_pr<br>ime | 376717<br>17       | 9               | 923                         | True                   | disrupted     | 0.3344709<br>897610921<br>3 | True                |                       |                       | Protein kinase domain | FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv   Protein Fusion: in frame {FBXL20:CDK12}               | NA                           |
| EVT-P-004<br>3910-T01-I<br>M6-88 | P-0043<br>910-T0<br>1-IM6 |            | FBXL2<br>0-CDK12  | FBXL20       | unknown      | thr<br>ee<br>_pr<br>ime | 376822<br>85       | 13              | 1159                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv   Protein Fusion: mid-exon {FBXL20:CDK12}                   | NA                           |
| EVT-P-003<br>6553-T01-I<br>M6-89 | P-0036<br>553-T0<br>1-IM6 |            | FBXL2<br>0-CDK12  | FBXL20       | unknown      | thr<br>ee<br>_pr<br>ime | 376190<br>62       | 1               | 246                         | False                  | retained      | 1.0                         | False               | Protein kinase domain |                       |                       | FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv   Protein Fusion: mid-exon {FBXL20:CDK12}                    | NA                           |
| EVT-P-004<br>8892-T01-I<br>M6-90 | P-0048<br>892-T0<br>1-IM6 |            | FBXL2<br>0-CDK12  | FBXL20       | unknown      | thr<br>ee<br>_pr<br>ime | 376824<br>32       | 13              | 1208                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv   Protein Fusion: mid-exon {FBXL20:CDK12}                   | NA                           |
| EVT-P-002<br>6288-T03-I<br>M6-91 | P-0026<br>288-T0<br>3-IM6 |            | FGFR4-CDK12       | FGFR4        | unknown      | thr<br>ee<br>_pr<br>ime | 376872<br>38       | 14              | 1381                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | FGFR4 (NM_213647) - CDK12 (NM_016507) rearrangement: t(5,17)(q35.2;q12)(chr5:g.176518227:chr17:g.37687238)   Protein Fusion: mid-exon {FGFR4:CDK12} | NA                           |
| EVT-P-006<br>0015-T01-I<br>M7-92 | P-0060<br>015-T0<br>1-IM7 |            | IGFBP4-CDK12      | IGFBP4       | unknown      | thr<br>ee<br>_pr<br>ime | 376872<br>28       | 14              | 1378                        | False                  | lost          | 0.0                         | False               |                       | Protein kinase domain |                       | IGFBP4 (NM_001552) - CDK12 (NM_016507) rearrangement: c.349+3777:IGFBP4_c.4132:CDK12dup   Protein Fusion: mid-exon {IGFBP4:CDK12}                   | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name         | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains     | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>8695-T01-I<br>M7-93  | P-0068<br>695-T0<br>1-IM7 |            | IKZF3-<br>CDK1<br>2 | IKZF3        | unknown      | threesome_prime | 37619043           | 1               | 240                         | False                  | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | IKZF3 (NM_012481) - CDK12 (NM_016507) rearrangement: c.8-3620:IKZF3_c.719:CDK12inv   Protein Fusion: mid-exon {IKZF3:CDK12}                           | NA                           |
| EVT-P-004<br>8104-T01-I<br>M6-94  | P-0048<br>104-T0<br>1-IM6 |            | MED13-<br>CDK12     | MED13        | unknown      | threesome_prime | 37627894           | 2               | 603                         | False                  | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | MED13 (NM_005121) - CDK12 (NM_016507) rearrangement: c.2264-2554:MED13_c.1809:CDK12inv   Protein Fusion: mid-exon {MED13:CDK12}                       | NA                           |
| EVT-P-006<br>5545-T01-I<br>M7-95  | P-0065<br>545-T0<br>1-IM7 |            | ODAD4-<br>CDK12     | ODAD4        | in-frame     | threesome_prime | 37648612           | 4               | 703                         | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | TTC25 (NM_031421) - CDK12 (NM_016507) rearrangement: c.1059-1499:TTC25_c.2109-392:CDK12dup   Protein Fusion: in frame {TTC25:CDK12}                   | NA                           |
| EVT-P-004<br>8892-T01-I<br>M6-97  | P-0048<br>892-T0<br>1-IM6 |            | SEC31A-<br>CDK12    | SEC31A       | unknown      | threesome_prime | 37682432           | 13              | 1208                        | False                  | lost          | 0.0                      | False               |                       | Protein kinase domain |                       | SEC31A (NM_014933) - CDK12 (NM_016507) rearrangement: t(4;17)(q21.22;q12)(chr4:g.83761890:chr17:g.37682432)   Protein Fusion: mid-exon {SEC31A:CDK12} | NA                           |
| EVT-P-000<br>5530-T02-I<br>M5-98  | P-0005<br>530-T0<br>2-IM5 |            | SYNRG-<br>CDK12     | SYNRG        | out-of-frame | threesome_prime | 37671800           | 9               | 923                         | True                   | disrupted     | 0.33447098976109213      | True                |                       |                       | Protein kinase domain | SYNRG (NM_007247) - CDK12 (NM_016507) rearrangement : c.118+961:SYNRG_c.2769-184:CDK12inv   Protein fusion: out of frame (SYNRG-CDK12)                | NA                           |
| EVT-P-004<br>1371-T01-I<br>M6-99  | P-0041<br>371-T0<br>1-IM6 |            | TBKB<br>P1-CDK12    | TBKB<br>P1   | in-frame     | threesome_prime | 37648975           | 4               | 703                         | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | TBKBP1 (NM_014726) - CDK12 (NM_016507) rearrangement: c.872+1376:TBKBP1_c.2109-29:CDK12dup   Protein Fusion: in frame {TBKBP1:CDK12}                  | NA                           |
| EVT-P-001<br>9776-T01-I<br>M6-100 | P-0019<br>776-T0<br>1-IM6 |            | USP6-<br>CDK12      | USP6         | in-frame     | threesome_prime | 37673592           | 10              | 949                         | True                   | disrupted     | 0.24573378839590443      | True                |                       |                       | Protein kinase domain | USP6 (NM_004505) - CDK12 (NM_016507) rearrangement: c.2916-470:USP6_c.2847-101:CDK12del   Protein Fusion: in frame {USP6:CDK12}                       | NA                           |
| EVT-P-005<br>9578-T01-I<br>M7-101 | P-0059<br>578-T0<br>1-IM7 |            | VMP1-<br>CDK12      | VMP1         | unknown      | threesome_prime | 37618564           | 1               | 80                          | False                  | retained      | 1.0                      | False               | Protein kinase domain |                       |                       | VMP1 (NM_030938) - CDK12 (NM_016507) rearrangement: c.582+1747:VMP1_c.240:CDK12dup   Protein Fusion: mid-exon {VMP1:CDK12}                            | NA                           |

# Figures

## domain retention outliers

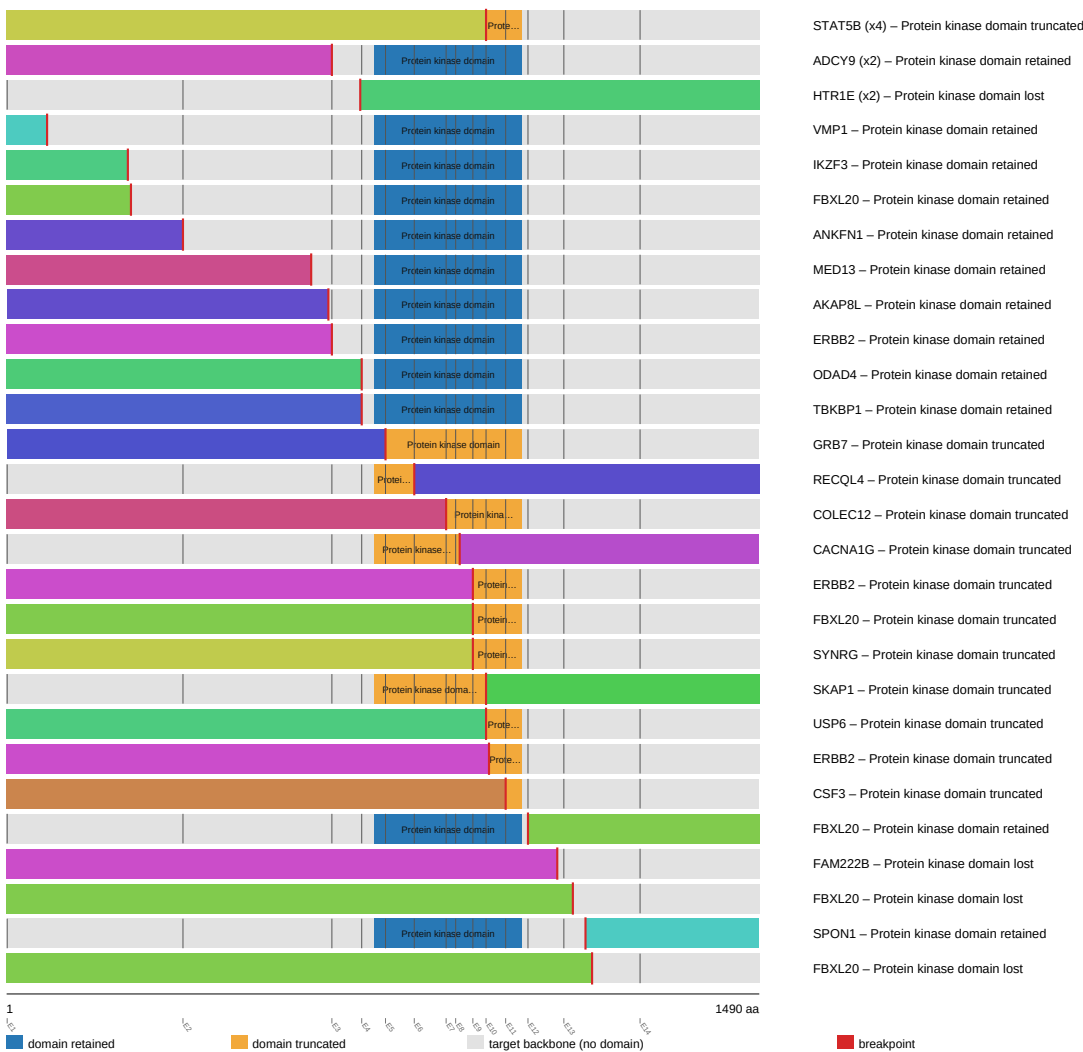
CDK12 domain-retention track



## fusion schematic

CDK12 fusion-transcript schematic

partner block → breakpoint → retained CDK12 portion (domain-colored, exon ticks)



Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk\_impact\_50k\_2026

